



Started on	Saturday, 12 September 2026, 9:57 PM
State	Finished
Completed on	Saturday, 12 September 2026, 9:58 PM
Time taken	1 min 26 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b	
s2		g	x	t	x	a	y	b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int main()
5  {
6      char s1[1000], s2[1000];
7
8      scanf("%s", s1);
9      scanf("%s", s2);
10
11     int m = strlen(s1);
12     int n = strlen(s2);
13
14     int dp[m + 1][n + 1];
15
16     for (int i = 0; i <= m; i++)
17         dp[i][0] = 0;
18
19     for (int j = 0; j <= n; j++)
20         dp[0][j] = 0;
21
22     for (int i = 1; i <= m; i++)
23     {
24         for (int j = 1; j <= n; j++)
25         {
26             if (s1[i - 1] == s2[j - 1])
27                 dp[i][j] = dp[i - 1][j - 1] + 1;
28             else
29             {
30                 if (dp[i - 1][j] > dp[i][j - 1])
31                     dp[i][j] = dp[i - 1][j];
32                 else
33                     dp[i][j] = dp[i][j - 1];
34             }
35         }
36     }
37
38     printf("%d", dp[m][n]);
39
40     return 0;
41 }
```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)